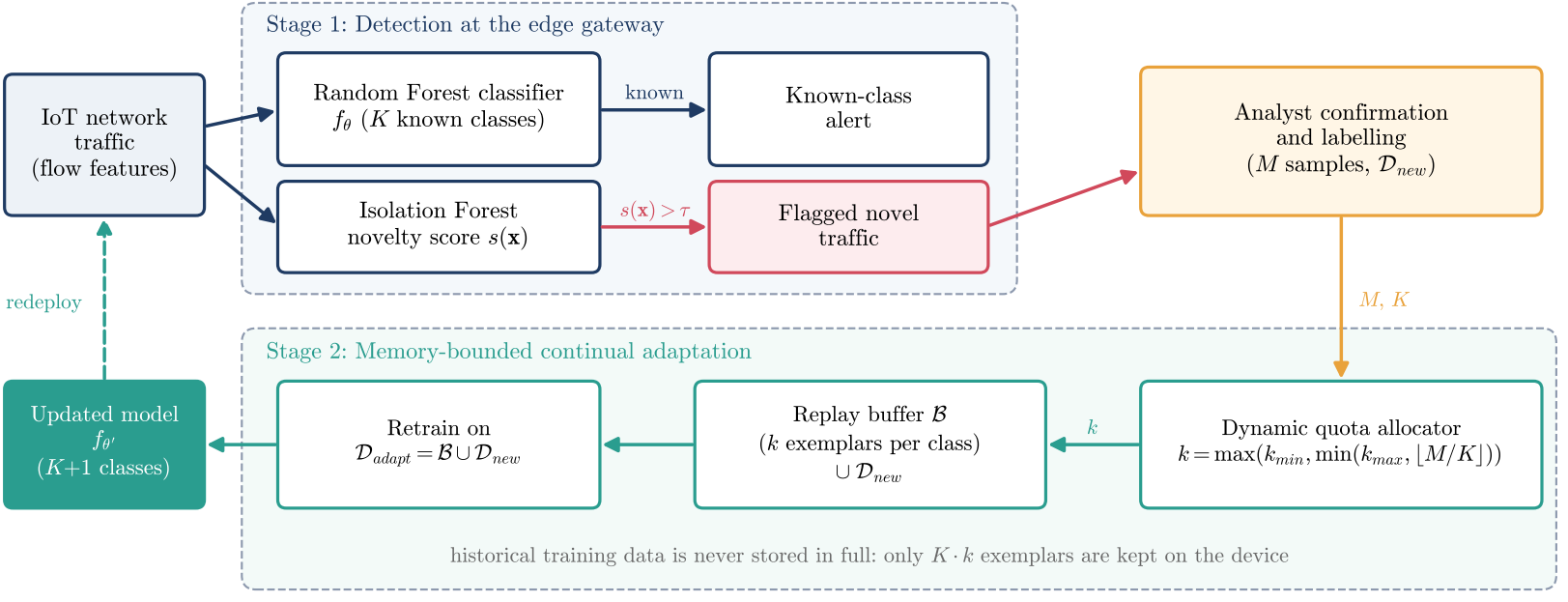
Stage 1: Detection at the edge gateway
IoT network traffic (flow features)
Random Forest classifier f_θ (K known classes)
known
Known-class alert
Isolation Forest novelty score $s(\mathbf{x})$
$s(\mathbf{x}) > \tau$
Flagged novel traffic
Analyst confirmation and labelling (M samples, $\mathcal{D}_{new}$)
M, K
redeploy
Stage 2: Memory-bounded continual adaptation
Updated model $f_{\theta'}$ (K+1 classes)
Retrain on $\mathcal{D}_{adapt} = \mathcal{B} \cup \mathcal{D}_{new}$
Replay buffer $\mathcal{B}$ (k exemplars per class) $\cup\, \mathcal{D}_{new}$
k
Dynamic quota allocator $k = \max(k_{min}, \min(k_{max}, \lfloor M/K \rfloor))$
historical training data is never stored in full: only $K \cdot k$ exemplars are kept on the device